

Numerical Simulation of ϕ^4 Scalar Field Dynamics in Noncommutative Spacetime

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Abstract

We simulate the non-local dynamics of a ϕ^4 scalar field in a 2D noncommutative spacetime. Using an FFT-based pseudo-spectral method, we evaluate the Moyal star product in momentum space and advance the field in time using a Velocity Verlet scheme. By simulating off-center collisions of Gaussian wave packets, we compare the noncommutative dynamics ($\theta \neq 0$) directly with the classical case ($\theta = 0$). Our results show that spacetime noncommutativity deforms the interaction geometry into a cross-like shape, creates structured interference patterns during overlap, and leaves behind a persistent phase shift after collision, all while preserving total energy to high precision.

1 Introduction

In theoretical physics, it is widely expected that classical spacetime geometry breaks down at very small distance scales near the Planck scale. One standard way to model these quantum spatial effects is noncommutative geometry, where normal coordinates are replaced by non-commuting operators satisfying

$$[\hat{x}^\mu, \hat{x}^\nu] = i\theta^{\mu\nu}, \quad (1)$$

where $\theta^{\mu\nu}$ is a constant antisymmetric matrix. On the space of functions, this noncommutativity replaces the usual pointwise multiplication with the Moyal star product, defined as

$$(f \star g)(x) = f(x) \exp\left(\frac{i}{2}\theta^{\mu\nu} \overleftarrow{\partial}_\mu \overrightarrow{\partial}_\nu\right) g(x). \quad (2)$$

In this project, we study the equation of motion for a scalar field with a ϕ^4 self-interaction in 2D noncommutative space. Due to the trace property of the Moyal product under spatial integrals, the kinetic and mass terms remain identical to the classical case, while the interaction term changes from ϕ^3 to the non-local product $\phi \star \phi \star \phi$.

To analyze these non-local dynamics, we developed a numerical simulation in Python. We use a spectral method based on the Fast Fourier Transform (FFT) to evaluate the star product efficiently in momentum space, along with a Velocity Verlet scheme for stable time integration. By simulating the off-center collision of Gaussian wave packets, we compare the noncommutative field ($\theta \neq 0$) directly with the classical field ($\theta = 0$). Furthermore, we monitor diagnostic quantities—such as energy conservation, peak amplitude, and momentum spectra—to confirm numerical stability and quantify the physical differences caused by spacetime noncommutativity.

2 Theoretical Framework

2.1 Noncommutative Spacetime and the Moyal Star Product

In noncommutative geometry, we treat spacetime coordinates as non-commuting operators that satisfy the fundamental commutation relation:

$$[\hat{x}^\mu, \hat{x}^\nu] = i\theta^{\mu\nu}, \quad (3)$$

where $\theta^{\mu\nu}$ is a constant antisymmetric matrix representing the noncommutativity scale. When $\theta^{\mu\nu} \rightarrow 0$, we get standard commutative spacetime back.

Instead of working with these operators directly, we can formulate field theories using standard functions on spacetime by replacing normal multiplication with the associative Moyal star product. For two smooth functions $f(x)$ and $g(x)$, it is defined as:

$$(f \star g)(x) = \exp \left(\frac{i}{2} \theta^{\mu\nu} \partial_{x^\mu} \partial_{y^\nu} \right) f(x)g(y) \Big|_{y=x}. \quad (4)$$

Expanding this exponential gives an infinite series of higher-order derivative terms:

$$(f \star g)(x) = f(x)g(x) + \frac{i}{2} \theta^{\mu\nu} (\partial_\mu f)(\partial_\nu g) - \frac{1}{8} \theta^{\mu\nu} \theta^{\rho\sigma} (\partial_\mu \partial_\rho f)(\partial_\nu \partial_\sigma g) + \mathcal{O}(\theta^3). \quad (5)$$

2.2 Fourier Representation

Calculating all these high-order spatial derivatives directly is numerically expensive. It is much easier to work in momentum space using the Fourier transform:

$$f(x) = \int \frac{d^d p}{(2\pi)^d} \tilde{f}(p) e^{ip \cdot x}, \quad g(y) = \int \frac{d^d k}{(2\pi)^d} \tilde{g}(k) e^{ik \cdot y}. \quad (6)$$

Substituting this into Eq. (4) gives:

$$(f \star g)(x) = \int \frac{d^d p}{(2\pi)^d} \frac{d^d k}{(2\pi)^d} \tilde{f}(p) \tilde{g}(k) \exp \left(\frac{i}{2} \theta^{\mu\nu} \partial_{x^\mu} \partial_{y^\nu} \right) e^{ip \cdot x} e^{ik \cdot y} \Big|_{y=x}. \quad (7)$$

Applying the derivatives to the plane waves yields $\partial_{x^\mu} \partial_{y^\nu} (e^{ip \cdot x} e^{ik \cdot y}) = -p_\mu k_\nu e^{ip \cdot x} e^{ik \cdot y}$. Setting $y = x$, this simplifies to:

$$(f \star g)(x) = \int \frac{d^d p}{(2\pi)^d} \frac{d^d k}{(2\pi)^d} \tilde{f}(p) \tilde{g}(k) \exp \left[i(p+k) \cdot x - \frac{i}{2} \theta^{\mu\nu} p_\mu k_\nu \right]. \quad (8)$$

In our 2D spatial grid, we have $[x, y] = i\theta$ (so $\theta^{xy} = \theta$ and $\theta^{yx} = -\theta$). The contraction becomes $\theta^{\mu\nu} p_\mu k_\nu = \theta(p_x k_y - p_y k_x)$, reducing the equation to:

$$(f \star g)(x) = \int \frac{d^2 p}{(2\pi)^2} \frac{d^2 k}{(2\pi)^2} \tilde{f}(p) \tilde{g}(k) \exp \left[i(p+k) \cdot x - \frac{i}{2} \theta (p_x k_y - p_y k_x) \right]. \quad (9)$$

This shows that spacetime noncommutativity is basically just a momentum-dependent phase shift in Fourier space.

2.3 Trace Property and Action Formulation

An important property of the Moyal product is its trace behavior under spacetime integration:

$$\int d^d x (f \star g)(x) = \int d^d x f(x)g(x). \quad (10)$$

To see why this works at first order in θ , we can look at the integral of the first derivative term from Eq. (5):

$$I = \frac{i}{2} \theta^{\mu\nu} \int d^d x (\partial_\mu f)(\partial_\nu g) = \frac{i}{2} \theta^{\mu\nu} \left[\int d^d x \partial_\mu (f \partial_\nu g) - \int d^d x f (\partial_\mu \partial_\nu g) \right]. \quad (11)$$

Assuming the fields vanish at infinity, the first total derivative term drops out. The second term is also zero because we are contracting an antisymmetric tensor ($\theta^{\mu\nu}$) with a symmetric double derivative ($\partial_\mu \partial_\nu$). So $I = 0$, which verifies Eq. (10).

Now, if we consider a real scalar field with a ϕ^4 interaction, the action is:

$$S_{\text{NC}} = \int d^4x \left[\frac{1}{2}(\partial_\mu \phi) \star (\partial^\mu \phi) - \frac{1}{2}m^2(\phi \star \phi) - \frac{\lambda}{4!}(\phi \star \phi \star \phi \star \phi) \right]. \quad (12)$$

Because of the trace property, the kinetic and mass terms are exactly the same as in normal commutative space. The noncommutativity only changes the non-linear interaction part.

2.4 Derivation of the Equation of Motion

If we vary the action with respect to ϕ and use the cyclic property under the integral, we get:

$$\delta S_{\text{NC}} = \int d^4x \left[-(\square + m^2)\phi - \frac{\lambda}{3!}(\phi \star \phi \star \phi) \right] \delta\phi = 0. \quad (13)$$

Setting $\delta S_{\text{NC}} = 0$ gives us the non-local equation of motion:

$$(\square + m^2)\phi + \frac{\lambda}{3!}(\phi \star \phi \star \phi) = 0. \quad (14)$$

This confirms that a free field travels normally, and θ only affects the non-linear interaction terms by making them non-local.

3 Numerical Methodology and Implementation

3.1 Spatial Discretization and Spectral Grid

To simulate the field $\phi(x, y, t)$, we use a 2D spatial grid $[-L/2, L/2] \times [-L/2, L/2]$ with $N \times N$ points and a resolution of $\Delta x = \Delta y = L/N$. With periodic boundary conditions, the grid points are:

$$x_i = -\frac{L}{2} + i\Delta x, \quad y_j = -\frac{L}{2} + j\Delta y, \quad (i, j = 0, 1, \dots, N-1). \quad (15)$$

We calculate spatial derivatives using a pseudo-spectral method based on the Fast Fourier Transform (FFT). The momentum grid coordinates (k_x, k_y) are:

$$k_x = \frac{2\pi}{L} \text{fttfreq}(N), \quad k_y = \frac{2\pi}{L} \text{fttfreq}(N). \quad (16)$$

In momentum space, the Laplacian is just a multiplication by $-K^2 = -(k_x^2 + k_y^2)$. We can then find the Laplacian in real space by applying the inverse FFT:

$$\nabla^2 \phi(x, y) = \mathcal{F}^{-1} \left\{ -K^2 \tilde{\phi}(k_x, k_y) \right\}. \quad (17)$$

3.2 Spectral Evaluation of the Moyal Star Product

As mentioned before, calculating the Moyal product in real space is slow. Instead, we do it in momentum space using discrete convolution. For two fields ϕ and ψ , their Moyal product in frequency space is:

$$\mathcal{F}\{\phi \star \psi\}(k_x, k_y) = \frac{1}{N^2} \sum_{p_x, p_y} \tilde{\phi}(k_x - p_x, k_y - p_y) \tilde{\psi}(p_x, p_y) \exp \left[-\frac{i}{2} \theta (k_x p_y - k_y p_x) \right]. \quad (18)$$

We get the spatial result by taking the inverse FFT and keeping the real part. To compute the cubic term $\phi \star \phi \star \phi$, we just apply this twice: first $\phi_2 = \phi \star \phi$, and then $\phi_3 = \phi_2 \star \phi$.

3.3 Time Integration Scheme: Velocity Verlet

To solve the equation of motion $\partial_t^2 \phi = \nabla^2 \phi - m^2 \phi - \frac{\lambda}{3!}(\phi \star \phi \star \phi)$, we rewrite it as two first-order equations:

$$\frac{\partial \phi}{\partial t} = v, \quad \frac{\partial v}{\partial t} = \nabla^2 \phi - m^2 \phi - \frac{\lambda}{3!}(\phi \star \phi \star \phi). \quad (19)$$

To keep the simulation stable and conserve energy over time, we use the Velocity Verlet algorithm. In our numerical implementation, we set $m = 0$ to isolate and study the purely non-local effects of the interaction term, simplifying the acceleration step to:

$$v\left(t + \frac{\Delta t}{2}\right) = v(t) + \frac{\Delta t}{2}a(t), \quad (20)$$

$$\phi(t + \Delta t) = \phi(t) + \Delta t v\left(t + \frac{\Delta t}{2}\right), \quad (21)$$

$$a(t + \Delta t) = \nabla^2 \phi(t + \Delta t) - \frac{\lambda}{6}(\phi \star \phi \star \phi)(t + \Delta t), \quad (22)$$

$$v(t + \Delta t) = v\left(t + \frac{\Delta t}{2}\right) + \frac{\Delta t}{2}a(t + \Delta t). \quad (23)$$

We run the classical ($\theta = 0$) and noncommutative ($\theta \neq 0$) simulations side-by-side with the exact same parameters so we can compare them directly.

3.4 Simulation Steps and Measurements

We divided the simulation workflow into three main parts:

1. **Static Operator Check:** Looking at how the Moyal product deforms the geometry of a static 2D Gaussian compared to a normal ϕ^3 term.
2. **Wave Packet Collision:** Simulating two Gaussian wave packets colliding off-center (with a non-zero impact parameter).
3. **Tracking Variables:** Measuring physical quantities over time to check the simulation, including:
 - **Peak Amplitude:** $\max |\phi(x, y, t)|$ to make sure the field doesn't blow up.
 - **Difference Norm:** $\sqrt{\langle (\phi_{\text{NC}} - \phi_{\text{C}})^2 \rangle}$ to see how much the non-local effects grow.
 - **Energy Drift:** Checking total energy to verify the integration scheme is stable.
 - **Momentum Spectrum:** Looking at the Fourier difference map $\log_{10} |\mathcal{F}\{\phi_{\text{NC}} - \phi_{\text{C}}\}|$ to see which spatial frequencies are affected most.

4 Results and Discussion

4.1 Static Non-local Deformation

Before looking at the time evolution, we first checked how noncommutativity changes the shape of the interaction term for a simple 2D Gaussian profile $\phi(x, y) = \exp(-(x^2 + y^2))$. Figure 1 compares the standard $\phi^3(x, y)$ term with the Moyal product $(\phi \star \phi \star \phi)(x, y)$ at $\theta = 1.5$.

As Figure 1 shows, the normal ϕ^3 term is completely symmetric and localized near the center. In contrast, the Moyal product spreads the field out and breaks the rotational symmetry into a cross-like shape. The phase shifts from the $\theta^{\mu\nu}$ matrix push the interaction strength into surrounding areas. This confirms that noncommutativity basically smears out the geometry of the interactions.

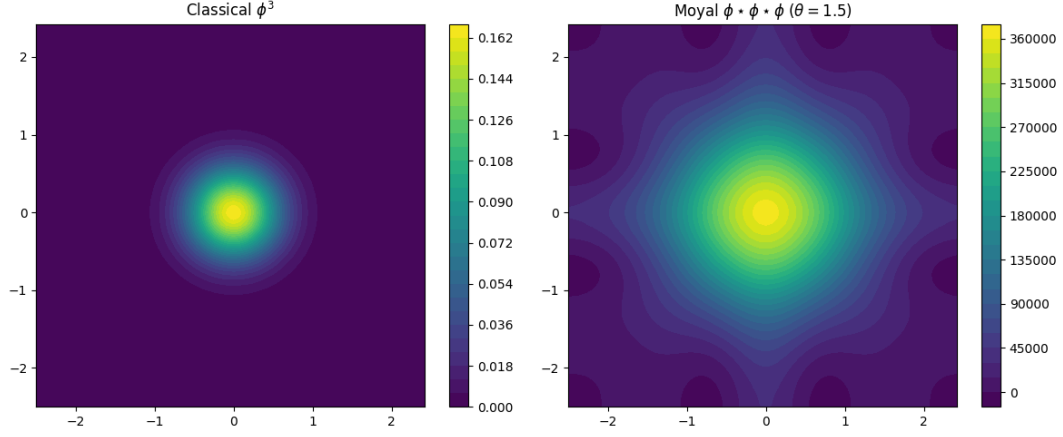


Figure 1: Spatial distribution of the interaction terms for a 2D Gaussian profile: standard ϕ^3 term (left) vs. the non-local Moyal product at $\theta = 1.5$ (right).

4.2 Wave Packet Collision Dynamics

Next, we simulated two Gaussian wave packets colliding off-center to see how the noncommutativity affects the dynamics. Figure 2 shows four snapshots of the simulation for both classical ($\theta = 0$) and noncommutative ($\theta = 5.0$) cases, along with the difference between them ($\Delta\phi$).

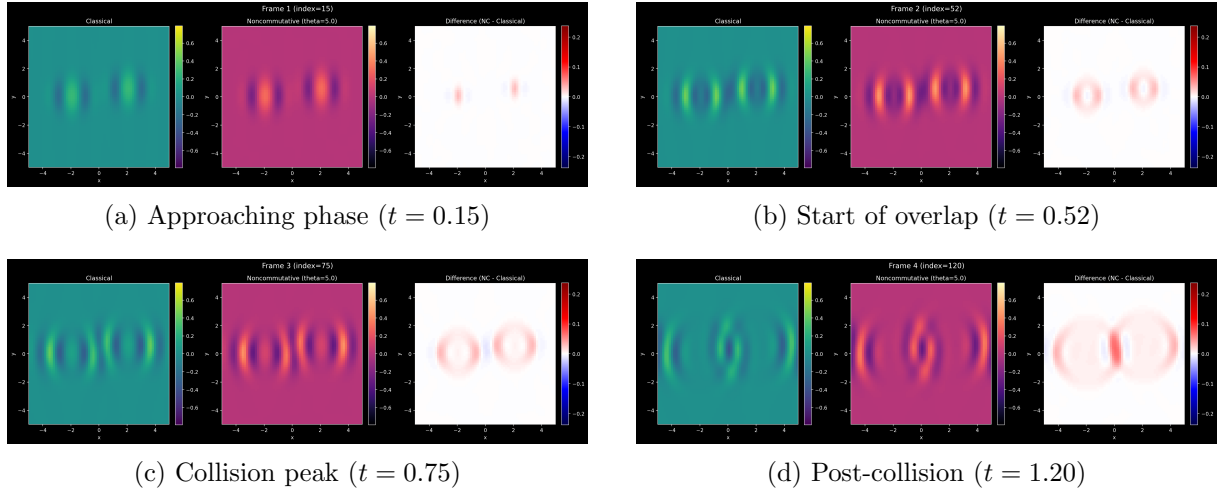


Figure 2: Snapshots of the collision. Left: classical field. Middle: noncommutative field ($\theta = 5.0$). Right: difference field $\phi_{\text{NC}} - \phi_{\text{C}}$.

We can see a few main stages in the collision:

1. **Before Overlap (Figure 2a):** While the packets are separated, the difference panel is basically zero. This confirms that θ doesn't affect free propagation.
2. **Initial Overlap (Figure 2b):** As the packets start to hit each other, the non-local terms kick in, and we start seeing small ring-like perturbations in the difference field.
3. **Maximum Collision (Figure 2c):** When the packets fully overlap, the non-linear self-interaction is at its strongest. We see clear, structured interference patterns in the difference panel.
4. **After Collision (Figure 2d):** Even after the wave packets pass through each other and move away, a residual phase shift remains in the center. The non-local interactions essentially leave a lasting structural mark on the field background.

4.3 Diagnostics and Stability

To make sure our numerical results are physically meaningful and stable, we tracked several quantities over time (Figure 3).

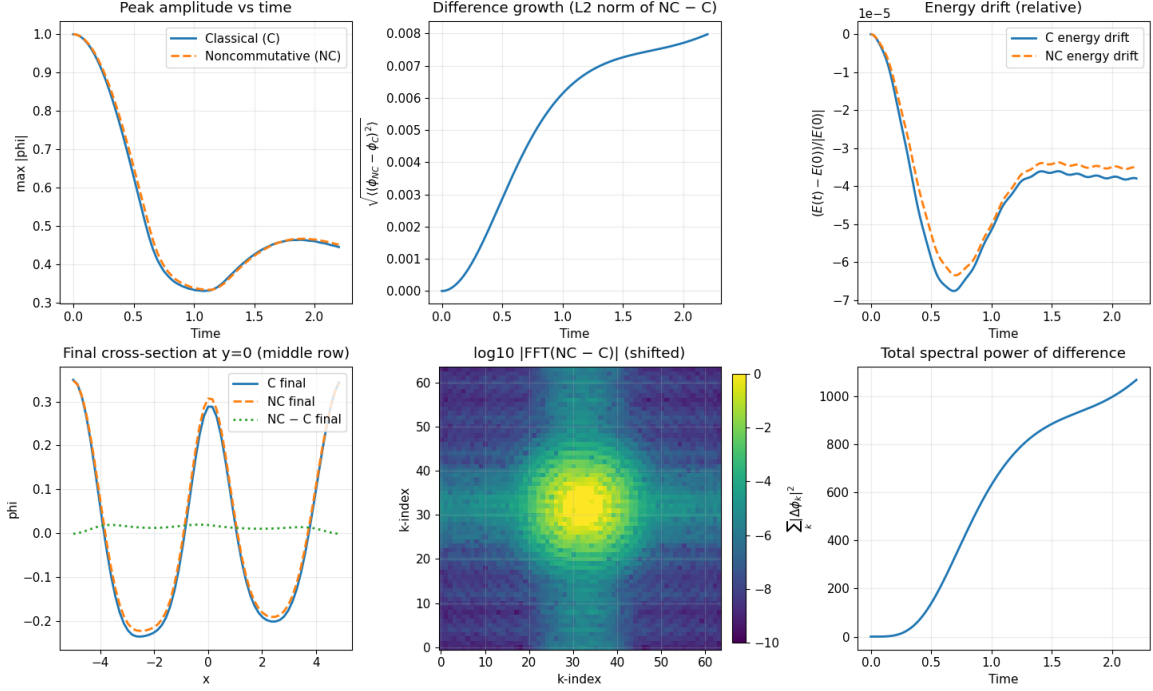


Figure 3: Measurements tracking classical and noncommutative ($\theta = 4.0$) dynamics: peak amplitude (top-left), difference norm (top-middle), energy drift (top-right), final 1D cross-section (bottom-left), Fourier power spectrum of the difference (bottom-middle), and integrated spectral power (bottom-right).

These plots tell us a few things:

- **Peak Amplitude (Top-Left):** The maximum field value drops as the waves disperse and then recovers slightly. The curves for $\theta = 0$ and $\theta = 4.0$ stay right on top of each other, meaning the noncommutativity doesn't cause any unphysical instabilities or blow-ups.
- **Difference Norm (Top-Middle):** The difference between the two fields grows smoothly and levels off after the collision, showing that the non-local effects accumulate normally.
- **Energy Conservation (Top-Right):** The energy drift for both runs stays around $\mathcal{O}(10^{-5})$ for the whole simulation. This proves that our Velocity Verlet setup is stable and that the differences we see are real physics, not just numerical errors.
- **Spectrum Shifts (Bottom-Middle & Right):** The momentum spectrum shows that the non-local modifications mostly happen at low-to-medium momenta (long wavelengths), and the total spectral power saturates smoothly after the collision.

5 Conclusion

In this project, we wrote a pseudo-spectral simulation to study a scalar field in a 2D noncommutative spacetime. By computing the Moyal product in momentum space using the Fast Fourier Transform (FFT), we were able to calculate the non-local interactions efficiently. We used a Velocity Verlet algorithm to solve the equation of motion, keeping the simulation stable over time.

Our results match the expected theoretical behavior:

- Free waves move exactly like they do in standard space; the noncommutativity only shows up when the fields interact.
- The Moyal product smears out the interaction geometry, breaking rotational symmetry.
- During wave packet collisions, the non-local effects create interference rings and leave behind a persistent phase shift after the waves scatter.
- The total energy remained highly stable (drift $\sim 10^{-5}$), confirming that our numerical method is reliable.

Overall, this shows that pseudo-spectral methods are a great tool for simulating Moyal field theories, and this code can be a good starting point for looking at more complex noncommutative setups in the future.

References

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